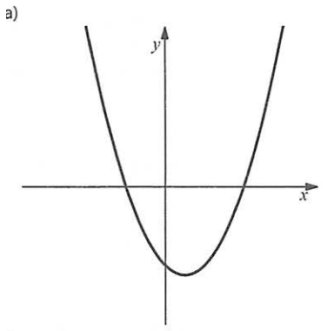
Further Algebra

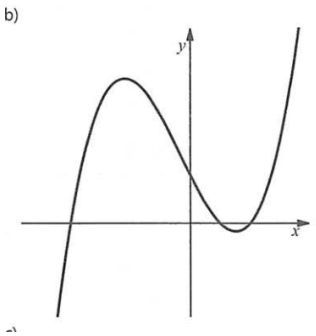
Questions 1–103

2.56 — Question 1. For each of the following graphs of $y = f(x)$, sketch the graph of $y = |f(x)|$:

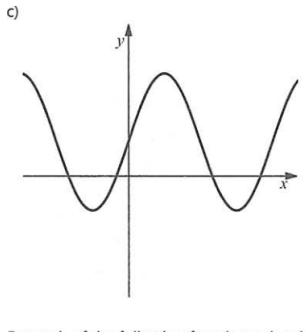
a)



b)



c)



2.56 — Question 2. For each of the following functions, sketch $y = |f(x)|$:

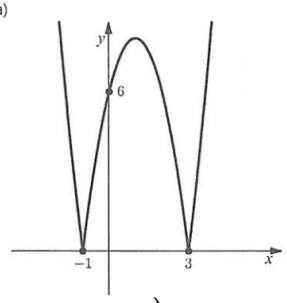
a) $f(x) = x^2 + x - 6$
b) $f(x) = x^2 + 7x + 10$
c) $f(x) = -x^2 + 2x + 8$
d) $f(x) = 6 + 4x - 2x^2$

2.56 — Question 3. Each of the following graphs are curves of the form $y = |ax^2 + bx + c|$ where a, b and c are integer constants to be found. In each case find the possible value(s) of a, b and c.

a)

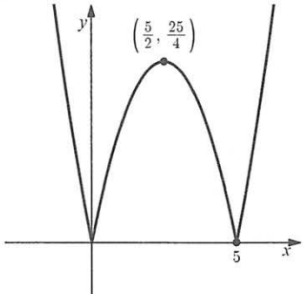
find the possible value(s) of a, b and c

a)



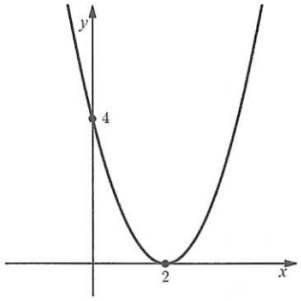
b)

b)



c)

c)

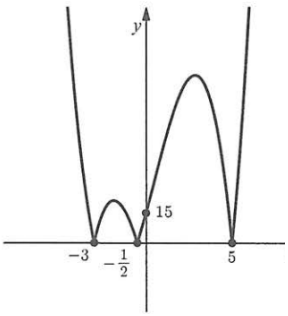


2.56 — Question 4. In each case, sketch $y = |f(x)|$:

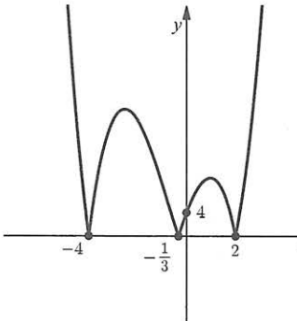
a) $f(x) = x^3 - 9x$
b) $f(x) = x^3 + x^2 - 12x$
c) $f(x) = x^3 - 7x + 6$
d) $f(x) = 14x - 5x^2 - x^3$

2.56 — Question 5. Each of the following graphs are curves of the form $y = |ax^3 + bx^2 + cx + d|$ where a, b, c and d are constants to be found. In each case find the possible value(s) of a, b, c and d.

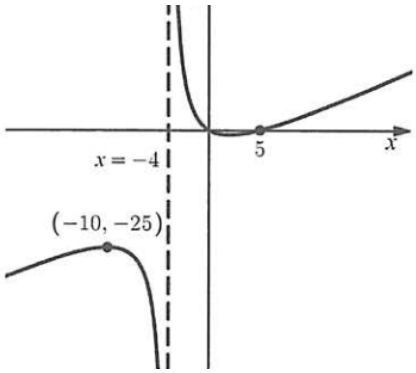
a)



b)



2.56 — Question 6. The diagram below shows part of the curve with equation $y = f(x)$.



Sketch $y = |f(x)|$.

2.56 — Question 7. In each case, sketch $y = |f(x)|$:

a) $f(x) = \dfrac{2x+3}{x-1}$

b) $f(x) = \dfrac{1-4x}{2x+5}$

2.56 — Question 8. In each case, sketch $y = |f(x)|$:

a) $f(x) = \ln(x-1)$

b) $f(x) = \sin x$ for $0 \leq x \leq 2\pi$

c) $f(x) = 1 - \cos x$ for $0 \leq x \leq 2\pi$

d) $f(x) = 1 - e^x$

e) $f(x) = \tan(2x)$ for $0 \leq x \leq \dfrac{\pi}{2}$

2.56 — Question 9. In each case, by drawing a suitable sketch find the number of solutions for each of the following:

a) $|\ln 2x| = 2x$

b) $\left|\frac{1}{x}\right| - 1 = x^2$

c) $|\sin x| = \frac{2}{\pi}x$

d) $|\cos x| = \frac{2}{\pi}x$

2.56 — Question 10. In each case, draw a suitable sketch to find the number of solutions for each of the following:

a) $|\sinh x| = 2$

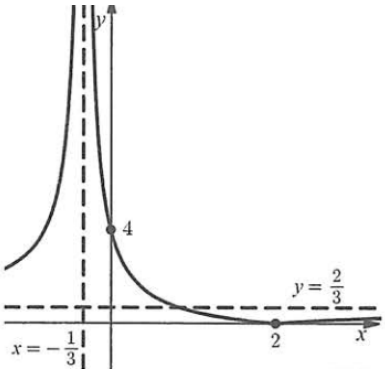
b) $2 - |\cosh x| = 1$

2.56 — Question 11. Solve each of the following:

a) $|\sinh x| = 1$

b) $|\cosh x - 2| = 1$

2.56 — Question 12. The curve below has equation $y = \left|\dfrac{ax+b}{cx+d}\right|$ where a, b, c and d are non-zero integers. Determine the values of a, b, c and d giving the values with the smallest possible integer coefficients.



2.57 — Question 1.

a) On the same set of axes (you do not need to label the points of intersection) sketch $y = |x^2 - 3x|$ and $y = \frac{5}{2} - \frac{1}{2}x$.

b) Hence solve $|x^2 - 3x| > \frac{5}{2} - \frac{1}{2}x$ giving your answer in interval notation.

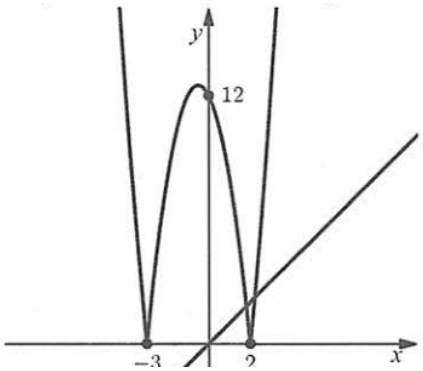
2.57 — Question 2. Solve each of the following, giving your answer in set notation:

a) $|x^2 - 4| > 3$

b) $|x^2 - 2x - 3| \leq 5$

c) $|x^2 - x - 6| > 2x$

2.57 — Question 3. The following graph shows the curves $y = |ax^2 + bx + c|$ and $y = x$, where a, b and c are non-zero integers.



a) Find the possible values of a, b and c.
b) Using your answers for part (a), determine the values for which $|ax^2 + bx + c| < x$ giving your answer in set notation.

2.57 — Question 4.

a) On the same set of axes (you do not need to label the points of intersection) sketch $y = |x^3 + x^2 - 4x - 4|$ and $y = 10x + 20$.
b) Hence solve $|x^3 + x^2 - 4x - 4| > 10x + 20$ giving your answer in interval notation.

2.57 — Question 5. Solve each of the following, giving your answer in set notation:

a) $|x^3 - x| > 3x$

b) $|x^3 - 2x| > x$

c) $|x^3 - 4x| \leq 3$

2.57 — Question 6.

a) On the same set of axes (you do not need to label the points of intersection) sketch $y = \left|\frac{2x+1}{x-1}\right|$ and $y = x + 3$.

b) Hence solve $\left|\frac{2x+1}{x-1}\right| > x + 3$ giving your answer in set notation.

2.57 — Question 7. Solve each of the following, giving your answer in set notation:

a) $\left|\frac{3x-2}{x+1}\right| > 2$

b) $\left|\frac{2x-1}{x-2}\right| < 3x + 2$

c) $\left|\frac{x^2-4}{x-1}\right| \leq 4$

2.57 — Question 8. Solve each of the following, giving your answer in interval notation:

a) $1 - |\cosh x| < 0$

b) $|1 - \sinh x| \leq 1$

c) $|\sinh x| + \cosh x \geq 2$

2.57 — Question 9. Given that $f(x) = |-1 + \ln x|$ for $x > 0$, find the values for which $f(x) \leq 2$. Give your answer in set notation.

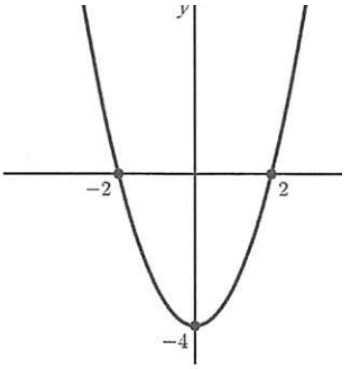
2.57 — Question 10. Given that $f(x) = \left|\frac{1}{|x|}\right|$, find the values for which $f(x) \geq 1$. Give your answer in set notation.

2.57 — Question 11. Given that $f(x) = \left|\cos x - \frac{1}{2}\right|$ for $-\pi < x \leq \pi$, find the values for which $f(x) \geq \frac{1}{2}$. Give your answer in interval notation.

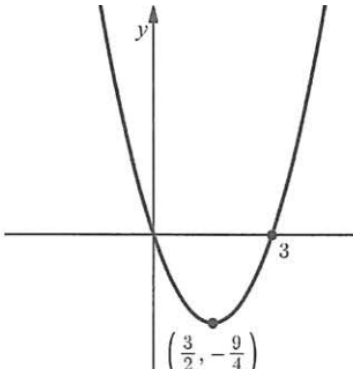
2.57 — Question 12. Given that $f(x) = \left|\frac{ax-1}{x-b}\right|$ for positive constants $1 < a < 4$ and $b > 1$, find the values for which $f(x) \geq 4$. Give your answer in set notation in terms of a and b.

2.58 — Question 1. For each of the following curves $y = f(x)$ shown below, sketch $y = \dfrac{1}{f(x)}$.

a)



b)



c)

2.58 — Question 2. Sketch each of the following, labelling any local maxima or minima with their coordinates:

a) $y = \dfrac{1}{3x - 1}$

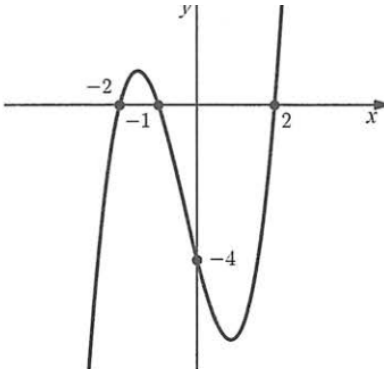
b) $y = \dfrac{1}{(x - 4)(x + 3)}$

c) $y = \dfrac{1}{2x^2 + 3x - 2}$

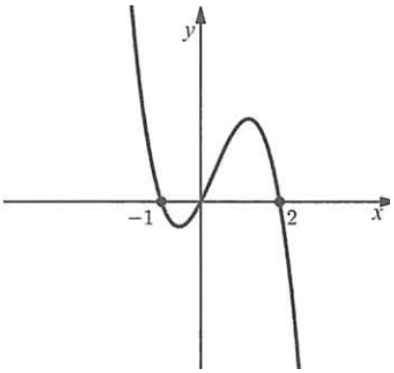
d) $y = \dfrac{1}{4 + 3x - x^2}$

2.58 — Question 3. For each of the following curves $y = f(x)$ shown below, sketch $y = \dfrac{1}{f(x)}$.

a)



b)



c)

2.58 — Question 4. Sketch each of the following:

a) $y = \dfrac{1}{x(x-1)(x-2)}$

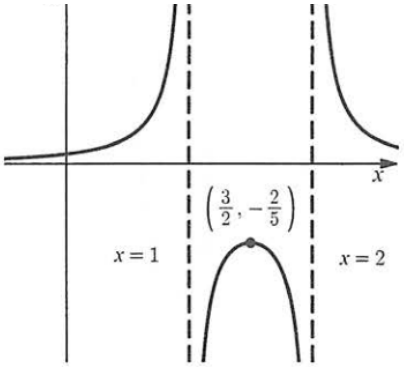
b) $y = \dfrac{1}{2x^3 - 9x^2 + 3x + 4}$

c) $y = \dfrac{1}{8 - 2x - 5x^2 - x^3}$

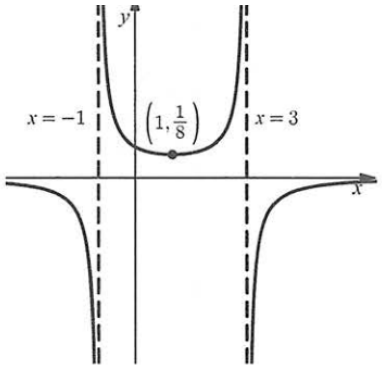
d) $y = \dfrac{1}{x^3 + x^2}$

2.58 — Question 5. Each of the following curves are of the form $y = \dfrac{1}{ax^2 + bx + c}$, where a, b and c are integers. Determine the values of a, b and c.

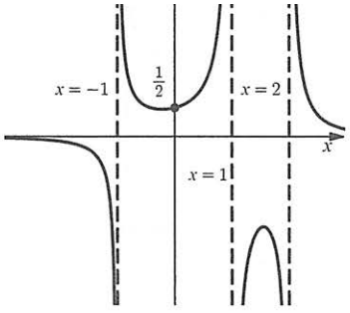
a)



b)



2.58 — Question 6. The following curve is of the form $y = \dfrac{1}{ax^3 + bx^2 + cx + d}$, where a, b, c and d are integers. Determine the values of a, b, c and d.



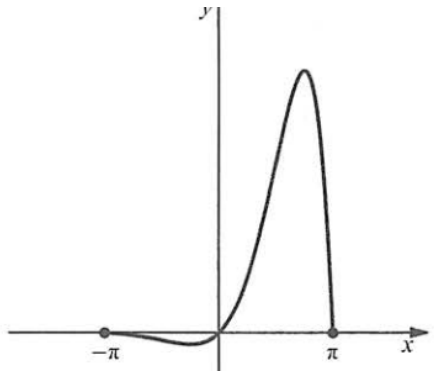
ketch the graph of $y = \frac{1}{f(x)}$ where

2.58 — Question 7. Sketch the graph of $y = \dfrac{1}{f(x)}$, where

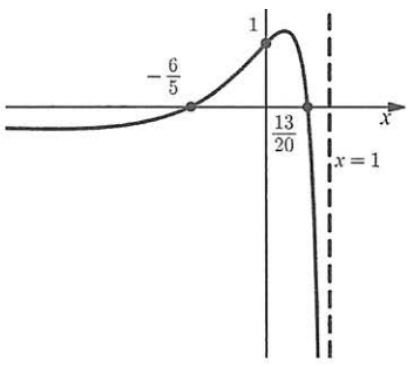
$$f(x) = \frac{x^2 + 4x}{x^2 - 4x - 5}.$$

2.58 — Question 8. For each of the following curves $y = f(x)$ shown below, sketch $y = \dfrac{1}{f(x)}$.

a)



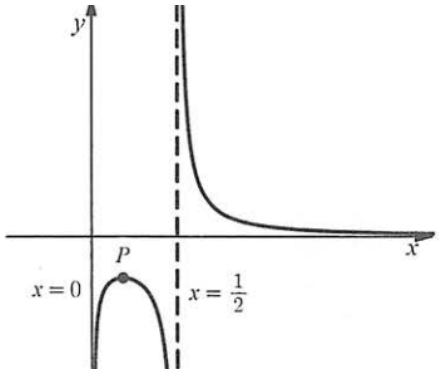
b)



2.58 — Question 9. The following curve is of the form

$$y = \frac{1}{ax\ln(bx)},$$

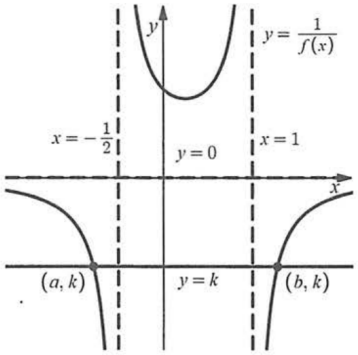
where a and b are non-zero integer constants to be found. The point P is at $\left(\frac{1}{2e}, -\frac{e}{4}\right)$. There are asymptotes at $x = 0$ and $x = \frac{1}{2}$. Determine the values of a and b.



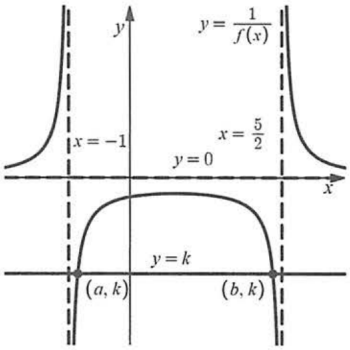
2.59 — Question 1. By first sketching a graph, find the values when $\frac{1}{x+2} \leq 0$, giving your answer in set notation.

2.59 — Question 2. Each of the following graphs show curves $y = \frac{1}{\mathrm{f}(x)}$ and the line $y = k$, for a constant k which intersect at the given points in terms of constants a and b
In each case, determine the values for which $\frac{1}{\mathrm{f}(x)} \leq k$, giving your answer in interval notation in terms of a and b

a)



b)



2.59 — Question 3. Solve each of the following equalities, giving your answer in interval notation:

a) $\frac{1}{x^2-4} \leq 0$

b) $\frac{1}{6+5x-x^2} < 0$

2.59 — Question 4. Solve each of the following equalities, giving your answer in set notation:

a) $\frac{1}{x^2-9} > \frac{1}{6}$

b) $\frac{1}{x(x-1)(x+4)} \leq 0$

2.59 — Question 5.

a) Show that $(4x-1)$ is a factor of

$$\mathrm{f}(x) = 12x^3 + x^2 - 9x + 2$$

b) Sketch $y = \frac{1}{12x^3+x^2-9x+2}$

c) Hence solve $\frac{1}{12x^3+x^2-9x+2} < 0$ giving your answer in set notation.

2.59 — Question 6.

a) Show that $(3x-1)$ is a factor of

$$\mathrm{f}(x) = 6x^3 - 5x^2 - 2x + 1$$

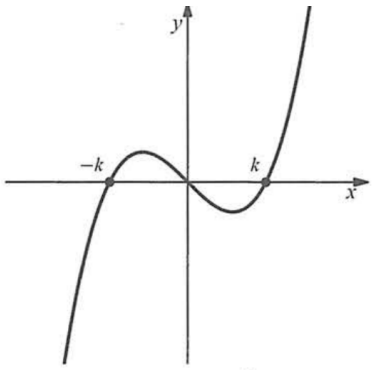
b) On the same set of axes sketch the $y = \frac{1}{6x^3-5x^2-2x+1}$ and $y = 1$
You do not need to label the points of intersection.

c) Hence solve $\frac{1}{6x^3-5x^2-2x+1} < 1$ giving your answer in interval notation.

2.59 — Question 7. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = x(x-k)(x+k)$$

where k is a positive integer is shown in the diagram below.



a) Sketch the graph of $y = \frac{1}{\mathrm{f}(x)}$

b) Hence, solve $\frac{1}{x(x-k)(x+k)} > 0$
Give your answer in set notation.

2.60 — Question 1. Find the equations of the asymptotes of each of the following curves:

a) $y = \dfrac{x^2 - x + 2}{x + 1}$

b) $y = \dfrac{x^2 + 2x - 6}{x - 4}$

c) $y = \dfrac{3x^2 + x - 7}{x + 2}$

d) $y = \dfrac{11 - x^2}{x + 3}$

2.60 — Question 2. Sketch the graphs of each of the following curves. You must clearly show any asymptotes:

a) $y = \dfrac{x^2 - 2x - 1}{x + 1}$

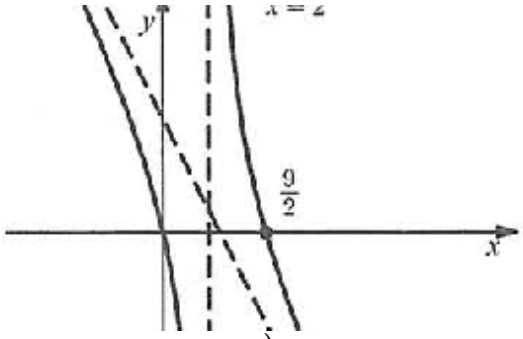
b) $y = \dfrac{2x^2 + 7x + 1}{x + 3}$

c) $y = \dfrac{6x^2 + x + 6}{1 - 3x}$

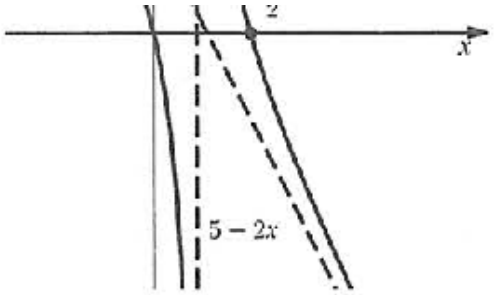
d) $y = \dfrac{51 + 5x - x^2}{x + 5}$

2.60 — Question 3. Each of the following curves are of the form $y = \dfrac{ax^2 + bx + c}{x + d}$, where a, b, c, d are integers. Determine the values of a, b, c, d.

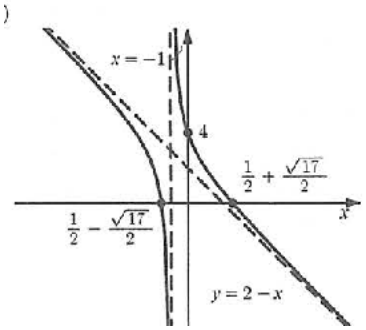
a)



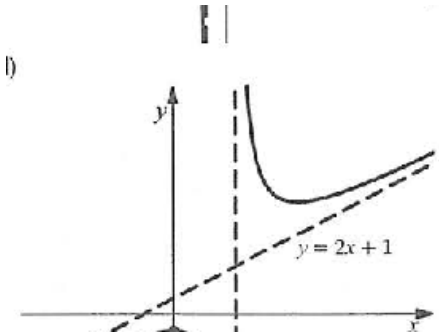
b)



c)



d)



2.60 — Question 4. Find the equations of the asymptotes of each of the following curves, where:

a) $y = \dfrac{ax^2}{x + 1}$ for non-zero constant a

b) $y = \dfrac{ax^2}{x - 2}$ for non-zero constant a

c) $y = \dfrac{ax^2 + bx}{x - 1}$ for non-zero constants a and b

d) $y = \dfrac{ax^2 + b}{x + a}$ for non-zero constants a and b

e) $y = \dfrac{ax^2}{bx - c}$ for non-zero constants a, b, c

2.60 — Question 5. One of the two asymptotes of the curve $y = \dfrac{x^2 + kx + 4}{x - 1}$, where k is a constant, is $y = x + 3$.

a) State the equation of the other asymptote.

b) Find the value of k.

2.60 — Question 6. One of the two asymptotes of the curve $y = \dfrac{2x^2 + kx - 3}{x - 2}$, where k is a constant, is $y = 2x + 7$.

a) State the equation of the other asymptote.

b) Find the value of k.

2.60 — Question 7. A curve has equation $y = \dfrac{ax^2 + bx + 4}{x - 1}$ and has asymptote $y = 3x + 2$.

a) Find the values of a and b.
b) Write down the equation of the other asymptote.
c) Sketch the curve.

2.60 — Question 8. The function f is defined where $x \neq 1$ by $f(x) = \dfrac{x^2 + 4x + 7}{x + 1}$. The line $y = k$ intersects the curve.

a) Show that $(k + 2)(k - 6) \geq 0$.
b) Hence, find the coordinates of the stationary points of the curve $y = f(x)$.
c) Find the equations of the asymptotes of the graph of $y = f(x)$.
d) Sketch the graph of $y = f(x)$.

2.60 — Question 9. The function f is defined where $x \neq -2$ by $f(x) = \dfrac{-x^2 + 6x + 12}{x + 2}$. The line $y = k$ intersects the curve.

a) Show that $(k - 6)(k - 14) \geq 0$.
b) Hence, find the coordinates of the stationary points of the curve $y = f(x)$.
c) Find the equations of the asymptotes of the graph of $y = f(x)$.
d) Sketch the graph of $y = f(x)$.

2.60 — Question 10. A new residential district is being developed. The rate of change of the population P (in thousands) is modelled by

$$\frac{dP}{dt} = \frac{2t^2 - t - 3}{t + 2}, \qquad t \geq 0,$$

where t is the time in years after construction begins.

a) Sketch the curve of $\dfrac{dP}{dt}$ for $t \geq 0$.
b) With reference to the rate of change of the population, explain what is happening to the population according to the model.
c) Find an expression for $P(t)$.
d) Given that initially there are 4 thousand residents, find the particular solution.
e) Determine the population after 4 years.

2.60 — Question 11. A curve has equation $y = \dfrac{ax^2 + bx + 7}{x + 3}$ and has asymptote $y = x + 1$.

a) Find the values of a and b.
b) Write down the equation of the other asymptote.
c) Without using Calculus, find the coordinates of the turning points.
d) Sketch the curve.

2.60 — Question 12. A curve has equation $y = \dfrac{x^2}{x + 2}$.

a) Find all the equations of the asymptotes of the curve.
b) Without using Calculus, find the coordinates of the stationary points of the curve.
c) Sketch the curve.

The region bounded by the curve, the x-axis and the lines $x = 0$ and $x = 2$ is rotated about the x-axis.

a) Show that the volume formed is given by $V = \pi\left(\dfrac{68}{3} - 32\ln 2\right)$ units^3.

2.60 — Question 13. A company manufactures wireless ear buds. The average production cost per batch C (in hundreds of pounds), when producing x batches per day, is modelled by

$$C = \frac{3x^2 + 4x + 2}{x + 1}.$$

a) Find all the equations of the asymptotes of the curve.
b) Without using Calculus, find the coordinates of the stationary points of the curve, giving your answers to three decimal places.
c) Sketch the curve.
d) Find the mean value of the cost function over the production interval $1 \leq x \leq 4$.
e) Interpret your answer in the context of the manufacturing process.

2.60 — Question 14. Find the equations of the asymptotes of each of the following curves:

a) $y = \dfrac{x^3 - 4}{x + 2}$
b) $y = \dfrac{x^3 + 3x^2 + 1}{x - 2}$
c) $y = \dfrac{1 - x^3}{x + 1}$

2.60 — Question 15. Sketch the graphs of each of the following curves. You must clearly show any asymptotes:

a) $y = \dfrac{x^3 + x^2 - 4}{x + 1}$

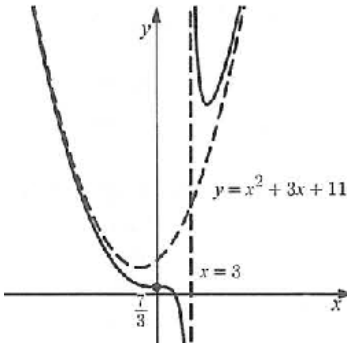
b) $y = \dfrac{18 + x - 3x^2 - x^3}{x + 3}$

c) $y = \dfrac{x^3 - 3x^2 - 4x + 12}{x + 1}$

d) $y = \dfrac{3x^3 + 4x^2 + x + 1}{x - 2}$

2.60 — Question 16. Each of the following curves are of the form $y = \dfrac{ax^3 + bx^2 + cx + d}{x + p}$, where a, b, c, d, p and q are integers. Determine the values of a, b, c, d, p.

a)



2.60 — Question 17. Find the equations of the asymptotes of each of the following curves, where:

a) $y = \dfrac{ax^3}{x+1}$ for non-zero constant a

b) $y = \dfrac{ax^3}{x-2}$ for non-zero constant a

c) $y = \dfrac{ax^3+bx^2}{x+1}$ for non-zero constants a and b

d) $y = \dfrac{ax^3+bx}{x^2+a}$ for non-zero constants a and b

e) $y = \dfrac{ax^3}{bx-c}$ for non-zero constants a, b, c

2.60 — Question 18. One of the two asymptotes of the curve $y = \dfrac{x^3+kx^2-2}{x-1}$, where k is a constant, is $y = x^2+5x+5$.

a) State the equation of the other asymptote.

b) Find the value of k.

2.60 — Question 19. One of the two asymptotes of the curve $y = \dfrac{x^3 - kx^2 + 47}{x + 3}$, where k is a constant, is $y = x^2 - 5x + 15$.

a) State the equation of the other asymptote.

b) Find the value of k.

2.61 — Question 1. The function f is defined by $f(x) = \dfrac{2x + 3}{x - 1}$.

a) State the domain of f.

b) State the equations of any asymptotes of the curve $y = f(x)$.

c) Find the range of f.

d) Sketch $y = f(x)$.

2.61 — Question 2. The function f is defined by $f(x) = \dfrac{3x-2}{4-3x}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Find the range of f.
d) Sketch $y = f(x)$.

2.61 — Question 3. The function f is defined by $f(x) = \dfrac{x^2-1}{x^2+4}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Without using Calculus, find the range of f.
d) Sketch $y = f(x)$.

2.61 — Question 4. The function f is defined by $f(x) = \dfrac{3x^2 - x - 2}{x^2 - x - 2}$.

a) Find the equations of all asymptotes of $y = f(x)$.
b) Without using Calculus, find the range of f.
c) Find the coordinates of the turning points.
d) Sketch the curve.

2.61 — Question 5. The function f is defined by $f(x) = \dfrac{x^2 - 1}{(x + 2)^2}$.

a) State the domain of f.
b) Find the equations of all asymptotes of $y = f(x)$.
c) Without using Calculus, determine whether the function has a maximum or minimum value.
d) Find the range of f.
e) Sketch $y = f(x)$.

2.61 — Question 6. Given that $f(x) = \dfrac{ax^2 + 6}{x^2 + 6x + 5}$, where a is a positive constant:

a) State the domain of f.

b) State the equations of any asymptotes of the curve $y = f(x)$ in terms of a.

Given that the horizontal asymptote is $y = 3$:

a) Find the y-coordinates of the turning points.

b) Find the range of f.

2.61 — Question 7. The function f is defined by $f(x) = \dfrac{2x^2 + 3}{x^2 + 1}$.

a) State the domain of f.

b) Without using Calculus, find the range of f.

c) Giving your answer in set notation, solve $f(x) \geq \dfrac{5}{2}$.

2.61 — Question 8. The function f is defined by $f(x) = \dfrac{x^2 - 6x + 16}{x^2 - 4}$.

a) State the domain of f.
b) Without using Calculus, find the range of f.
c) Sketch the curve $y = f(x)$.
d) Solve the inequality $-4 \leq f(x) < 1$.

2.61 — Question 9. For constant a, the function f is defined by $f(x) = \dfrac{x^2 + ax + 4}{x^2 + 1}$.

a) Show that if $f(x) \geq 0$, then $a^2 \leq 16$.
b) For $a = 4$, without using Calculus, find the range of f.
c) For $a = 4$, solve $f(x) \leq 2$.

2.61 — Question 10. Two curves C_1 and C_2 are defined by

$$C_1 : y = \frac{6}{x^2 - 5x + 6}, \qquad C_2 : y^2 = \frac{6}{x^2 - 5x + 6}.$$

a) Sketch C_1.
b) Determine the exact coordinates, if they exist, at which C_1 and C_2 intersect.
c) Hence solve $\dfrac{6}{x^2 - 5x + 6} > \sqrt{\dfrac{6}{x^2 - 5x + 6}}$.

2.61 — Question 11. The function f is defined by $f(x) = \dfrac{x^2 + 1}{x}$.

a) State the equations of any asymptotes of the curve $y = f(x)$.
b) Without using Calculus, find the range of f.
c) Solve $f(x) > x$.

2.61 — Question 12. The function f is defined by $f(x) = \dfrac{x^2 - 3}{x + 2}$.

a) State the equations of any asymptotes of the curve $y = f(x)$.
b) Without using Calculus, find the range of f.
c) Solve $f(x) \leq x - 2$.

2.61 — Question 13. The function f is defined by $f(x) = \dfrac{3x^2 - 2x + 19}{x - 2}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Without using Calculus, find the range of f.

2.61 — Question 14. The function f is defined by $f(x) = \dfrac{2x^2 + 5x - 3}{x + 1}$.

a) State the domain of f.
b) State the equations of any asymptotes of the curve $y = f(x)$.
c) Without using Calculus, find the range of f.

2.61 — Question 15. Given that $f(x) = \dfrac{x^2 + ax + 1}{x + 4}$, where a is a constant:

a) State the domain of f.
b) Find the value of a such that the graph has an oblique asymptote of $y = x$.
c) For the value of a found in part (b), without using Calculus, find the range of f.

2.61 — Question 16. Given that $f(x) = \dfrac{x^2 + ax + 4}{x - 3}$, where a is a constant:

a) Find the equation of the oblique asymptote in terms of a.

b) Find the value(s) of a such that the oblique asymptote passes through the point $(0, -1)$.

For the value of a found in part (b):

a) Find the domain of f.

b) Without using Calculus, find the range of f.

c) Sketch $y = f(x)$.

2.61 — Question 17. The function f is defined by $f(x) = \dfrac{2x^2 + x - 6}{x - 2}$.

a) State the equations of any asymptotes of the curve $y = f(x)$.

b) Without using Calculus, find the range of f.

c) Solve $f(x) > 2x + 6$.

2.61 — Question 18. The function f is defined by $f(x) = \dfrac{x^2 + ax - 1}{x + 1}$, where a is a constant.

a) State the vertical asymptote and find the equation of the oblique asymptote in terms of a.

b) Find the value of a such that $f(2) = \dfrac{1}{3}$.

c) For the value of a found in part (b), without using Calculus, find the range of f.

d) For the value of a found in part (b), solve $f(x) \geq x + 1$.

2.61 — Question 19. Two curves C_1 and C_2 are defined by $y = f(x)$ and $y = g(x)$ respectively, where

$$f(x) = \frac{x^2 - 2x + 5}{x - 1}, \qquad g(x) = \big(f(x)\big)^2.$$

a) State the equations of any asymptotes of the curve $y = f(x)$.

b) Without using Calculus, find the range of f.

c) Sketch the curve C_1, including any turning points on your diagram.

d) Determine the exact values, if they exist, at which C_1 and C_2 intersect.

e) Hence solve $\dfrac{x^2 - 2x + 5}{x - 1} > \sqrt{\dfrac{x^2 - 2x + 5}{x - 1}}$.

2.62 — Question 1. Write down the equation of the image of the ellipse given by

$$\frac{x^2}{3} + \frac{y^2}{4} = 1$$

when it has been transformed by each of the following transformations, giving your answer in the form

$$\frac{(x-p)^2}{a} + \frac{(y-q)^2}{b} = 1,$$

where a, b, p and q are constants to be found.

a) Rotated by $\frac{\pi}{2}$ radians anti-clockwise about $(0,0)$ followed by an enlargement of scale factor 2 with centre $(0,0)$.

b) A reflection in the x-axis, followed by a rotation by $\frac{\pi}{2}$ radians anti-clockwise about $(0,0)$.

c) A translation by vector $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, followed by a reflection in the line $y = -x$.

d) Rotated by $\frac{3\pi}{2}$ radians anti-clockwise about $(0,0)$ followed by an enlargement of scale factor $\frac{1}{2}$ with centre $(0,0)$.

e) Stretch parallel to the x-axis, scale factor 3, followed by a translation by vector $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$, followed by a rotation of $\frac{3\pi}{2}$ radians anti-clockwise about $(0,0)$.

2.62 — Question 2. Write down the equation of the image of the hyperbola given by

$$\frac{x^2}{6} - \frac{y^2}{3} = 1$$

when it has been transformed by each of the following transformations:

a) Rotated by $\frac{3\pi}{2}$ radians anti-clockwise about $(0,0)$ followed by a stretch parallel to the y-axis, scale factor 4.

b) A reflection in the y-axis, followed by a rotation by $\frac{\pi}{2}$ radians anti-clockwise about $(0,0)$.

c) A translation by vector $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$, followed by a reflection in the line $y = x$.

d) Stretch parallel to the x-axis, scale factor $\frac{1}{3}$, followed by a reflection in $y = -x$.

e) Enlargement of scale factor 2 with centre $(0,0)$, followed by a translation by vector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, followed by a rotation of $\frac{\pi}{2}$ radians anti-clockwise about $(0,0)$.

2.62 — Question 3. Write down the equation of the image of the rectangular hyperbola given by $xy = 16$ when it has been transformed by each of the following transformations:

a) Rotated by $\frac{3\pi}{2}$ radians anti-clockwise about $(0, 0)$ followed by an enlargement of scale factor 3 with centre $(0, 0)$.

b) A reflection in the x-axis, followed by a rotation by $\frac{\pi}{2}$ radians anti-clockwise about $(0, 0)$.

c) A translation by vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, followed by a reflection in the line $y = -x$.

d) Stretch parallel to the x-axis, scale factor 2, followed by a translation by vector $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$.

e) Enlargement of scale factor 2 with centre $(0, 0)$, followed by a translation by vector $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$, followed by a rotation of $\frac{3\pi}{2}$ radians anti-clockwise about $(0, 0)$.

2.62 — Question 4. Each of the following curves are transformed by a stretch parallel to the y-axis scale factor 2, followed by a translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$, followed by a rotation of π radians anti-clockwise about $(0, 0)$. Find the equation of the image:

a) $\frac{x^2}{5} + \frac{y^2}{12} = 1$

b) $\frac{x^2}{4} - \frac{y^2}{6} = 1$

c) $xy = 11$

d) $x^2 - y^2 = 9$

2.62 — Question 5. Describe a pair of transformations that map:

a) $x^2 + 9y^2 = 1$ to $\frac{(x-2)^2}{36} + \frac{y^2}{4} = 1$

b) $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to $\frac{(x-3)^2}{9} - \frac{y^2}{16} = 1$

c) $xy = 1$ to $(x-2)y = 3$

d) $\frac{x^2}{16} + \frac{y^2}{9} = 1$ to $\frac{(x-1)^2}{16} + \frac{(y+3)^2}{36} = 1$

e) $\frac{x^2}{4} - \frac{y^2}{5} = 1$ to $\frac{(2-x)^2}{8} - \frac{y^2}{18} = 1$

f) $\frac{x^2}{25} + \frac{y^2}{4} = 1$ to $\frac{(x+2)^2}{4} + \frac{(y-1)^2}{25} = 1$

g) $xy = 1$ to $(x+1)(y-2) = 4$

h) $\frac{y^2}{16} - \frac{x^2}{9} = 1$ to $9(y-1)^2 - 4x^2 = 144$

i) $\frac{x^2}{5} + \frac{y^2}{3} = 1$ to $x^2 + 5(y^2 - 4y) = -5$

j) $x^2 - \frac{y^2}{5} = 1$ to $x^2 = 5(y^2 + 2x)$

2.62 — Question 6. The curve C_1, given by equation $\frac{x^2}{2} + \frac{y^2}{4} = 1$, is mapped to curve C_2 with equation $2(x-1)^2 + (y+3)^2 = 16$.

a) Describe the transformations that map curve C_1 to curve C_2.

b) Find where the curve C_2 intersects with the coordinate axes.

c) Hence, sketch the curve C_2.

2.62 — Question 7. A curve C has equation $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a) State the type of curve C is.

b) State the asymptotes of C.

c) Given that C is translated by vector $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ followed by a reflection in the line $y = -x$, find the asymptotes of the image of C.

2.62 — Question 8. The curve $x^2 + 4y^2 = 1$ is transformed by the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ followed by $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

a) Describe the geometrical transformations represented by matrix $M = AB$.

b) Find the equation of the image of the ellipse under the transformation given by matrix M.

2.62 — Question 9. The curve $x^2 - y^2 = 4$ is transformed by the matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ followed by $B = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$.

a) Describe the geometrical transformations represented by matrix $M = AB$.

b) Find the equation of the image of the curve under the transformation given by matrix M.

2.62 — Question 10. A curve C has equation $\frac{x^2}{9} - \frac{y^2}{4} = 1$. The curve C is transformed onto curve C' by the transformation represented by matrix A. In each of the following cases, find the equations of the asymptotes of curve C':

a) $A = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$

b) $A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$

c) $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & -1 \\ 3 & 1 \end{pmatrix}$

d) $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$

2.62 — Question 11. The ellipse E given by equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is transformed to an ellipse E' by the matrix $M = \frac{1}{3}\begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$.

a) Find the determinant of M.
b) Find the equation of the ellipse E'.
c) It is known that the area enclosed by the ellipse E is 6π. Find the area enclosed by the curve E'.

2.62 — Question 12. The curve C given by equation $x^2 + y^2 = 9$ is transformed to curve C' by the matrix $M = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$.

a) Find the determinant of M.
b) Show that the equation of the curve C' is given by $ax^2 - bxy + cy^2 = 81$, where a, b and c are constants to be found.
c) Find the area enclosed by the curve C'.

2.62 — Question 13. The curve C given by equation $4x^2 - 8xy + 8y^2 = 1$ is transformed to curve C' by the matrix $M = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$.

a) Find the equation of the curve C'.
b) Find the area enclosed by the curve C'.
c) Hence, find the area enclosed by the curve C.

2.62 — Question 14. The curve C_1, given by equation $\frac{x^2}{16} - \frac{y^2}{4} = 1$, is mapped to curve C_2 with equation $x^2 + 8x - y^2 = 0$.

a) State the equations of the asymptotes of curve C_1.
b) Describe the transformations that map curve C_1 to curve C_2.
c) Determine the equations of the asymptotes of curve C_2, giving your answer in the form $y = mx + c$, where m and c are constants to be found.
d) Find where the curve C_2 intersects with the coordinate axes.
e) Hence, sketch the curve C_2.

2.62 — Question 15. A curve C has equation $\frac{x^2}{9} + \frac{y^2}{16} = 1$.

a) Sketch the curve C.

b) State two transformations that map C onto a circle of radius 1 with centre $(0, 0)$.

c) Find the area enclosed by the curve C.

The curve C is translated by vector $\begin{pmatrix} p \\ q \end{pmatrix}$, where p and q are positive constants, to form a new curve D. The curve D has a single point of contact with both the x-axis and the y-axis.

d) Find the values of p and q.

The curve C is stretched parallel to the x-axis by scale factor 2 to form curve E.

e) Find the equation of the transformed curve E in the form $ax^2 + by^2 = c$, where a, b and c are integers.

2.62 — Question 16. The curve C_1, given by equation $\frac{x^2}{16} + y^2 = 1$, has two transformations applied to it followed by a translation to curve C_2 with equation $6x^2 + 24x + 2y + y^2 = -19$.

a) Find where the curve C_1 intersects with the coordinate axes.

b) Describe the transformations that map curve C_1 to curve C_2.

c) Find where the curve C_2 intersects with the coordinate axes.

d) By using transformations, find the values of the maximum and minimum values of x and y on C_2.

e) Hence, sketch the curve C_2.

2.62 — Question 17. The curve C_1 has equation $\frac{x^2}{4} - \frac{y^2}{25} = 1$.

a) Sketch the curve C_1.

The curve C_1 is translated by vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$, where k is a positive constant, followed by a reflection in the line $y = -x$, to give curve C_2. The curve C_2 passes through the origin.

b) Determine the value of k.

c) Find the equations of the asymptotes of C_2, giving your answers in the form $y = mx + c$, where m and c are constants to be found.

2.62 — Question 18. A hyperbola has equation $\frac{x^2}{4} - \frac{y^2}{3} = 1$. The hyperbola is transformed as follows: translated by vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$, followed by a rotation anti-clockwise by $\frac{\pi}{2}$, followed by a stretch in the x-direction by factor 3.

a) Show that the transformed hyperbola can be written in the form $4x^2 - 27y^2 + ax + by + c = 0$, where a, b and c are constants to be found.

b) Given that the line $y = k$ intersects the new hyperbola at two points, find the possible values of k, giving your answer in set notation.

2.62 — Question 19. A hyperbola H has equation $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a) Show that the line with equation $4x - 3y = 11$ intersects H at only one point.

The hyperbola H is translated by the vector $\begin{pmatrix} a \\ -1 \end{pmatrix}$ to form a new curve K. The line $4x - 3y = 11$ is now an asymptote to the translated curve K.

b) Find the value of a.

2.62 — Question 20. The curve C_1 has equation $\frac{x^2}{25} + \frac{y^2}{4} = 1$. The curve C_2 is the image of C_1 after a reflection across the line $y = x$, followed by a translation by vector $\begin{pmatrix} a \\ b \end{pmatrix}$ for positive constants a and b, followed by an enlargement of scale factor 2 about $(0, 0)$. C_2 touches the x-axis at a single positive value of x and it touches the y-axis at a single positive value of y.

a) Find the value of a and b.

b) Find the equation of C_2 for your values of a and b, giving your answer in the form $\frac{(x-p)^2}{r} + \frac{(y-q)^2}{s} = 1$, where p, q, r and s are constants to be found.

2.62 — Question 21. The curve C_1 has equation $\frac{(x-2)^2}{4} + \frac{y^2}{9} = 1$. The curve C_2 is the image of C_1 after a translation by vector $\begin{pmatrix} a \\ b \end{pmatrix}$ for non-zero negative constants a and b, followed by a reflection in the y-axis. Curve C_2 touches the curve C_1 at a single point and also touches an axis once.

a) State the name given to the curve C_1.
b) Find the values of a and b.
c) Find the equation of C_2.
d) State the point of intersection of the curves C_1 and C_2.
e) Sketch both C_1 and C_2 on the same axes.
f) Find the equations of the tangent lines that are a tangent line to both C_1 and C_2.

2.62 — Question 22. An ellipse E has equation $\frac{x^2}{4} + \frac{y^2}{2} = 1$.

a) Write down the coordinates of the points where the ellipse E intersects the coordinate axes.

The ellipse E is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$, where k is a positive constant, followed by an enlargement of scale factor $\sqrt{3}$ about the origin.

b) Find the equation of the transformed ellipse in the form $\frac{(x-a)^2}{p} + \frac{(y-b)^2}{q} = 1$, where a, b, p and q are constants to be found, giving your answer in terms of k.

A straight line L has equation $y = -\frac{1}{2}x + 5$.

c) Show that if the transformed ellipse intersects the line L then the points of intersection must satisfy $3x^2 - (4\sqrt{3} + 20)x + 6k^2 + 72 = 0$.
d) Given that the line L is a tangent to the translated ellipse, find the possible values of k.

2.62 — Question 23. A hyperbola H has equation $\frac{x^2}{25} - \frac{y^2}{4} = 1$.

a) Sketch the hyperbola H.

It is known that the line with equation $y = x + k$ for constant k intersects the hyperbola H at two distinct points.

b) Show that $k > \sqrt{21}$ and $k < -\sqrt{21}$.

H is translated by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$, where a and b are non-zero constants, to form another hyperbola whose equation is $4x^2 - 25y^2 - 32x + 50y = c$.

c) Find the values of the constants a, b and c.

d) Hence find an equation for each of the two tangents to $4x^2 - 25y^2 - 32x + 50y = c$ that are parallel to the line $y = x$.

2.62 — Question 24. An ellipse E has equation $\frac{x^2}{4} + \frac{y^2}{100} = 1$. The line with equation $y = 2x + k$ for constant k intersects E at two distinct points.

a) Show that $-\sqrt{116} < k < \sqrt{116}$.

E is translated by vector $\begin{pmatrix} a \\ b \end{pmatrix}$, where a and b are non-zero constants, to form a curve with equation $25x^2 + y^2 - 150x + 10y = c$.

b) Find the values of the constants a, b and c.

c) Hence find an equation for each of the two tangents to $25x^2 + y^2 - 150x + 10y = c$ that are parallel to the line $y = 2x$.

2.62 — Question 25. A curve C_1 has equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a) State the name given to the curve C_1.

b) Sketch the curve C_1.

The curve C_1 is transformed onto curve C_2 by a combination of a translation followed by a stretch. The equations of the asymptotes of C_2 are $y = 3x - k$ and $y = k - 3x$, for positive constant k.

c) In the case that $k = 6$, find the equation of C_2.

d) Determine the value of k such that C_1 and C_2 intersect at exactly three points.